THE DERIVATIVE · TOPIC 1

Differentiating sin, cos, tan, e^x and $\ln x$

Thirty questions on the trigonometric, exponential and logarithmic derivatives, in three tiers of ten.

Pure Mathematics 1, 7.3–7.4 · Pure Mathematics 2, 4.1–4.4 · 30 questions · 60 marks

KEY WORDS

- radian
- chain rule
- product rule
- quotient rule
- natural logarithm
- exponential function
- logarithmic differentiation
- stationary point
- second derivative

REMEMBER

1. $(\sin x)' = \cos x$, $(\cos x)' = -\sin x$, $(\tan x)' = \sec^2 x$.
2. $(e^x)' = e^x$ and $(\ln x)' = \dfrac{1}{x}$.
3. Chain rule: differentiate the outside, then multiply by the derivative of the inside.
4. $\dfrac{d}{dx}\ln f(x) = \dfrac{f'(x)}{f(x)}$.
5. Radians throughout. Show the rule you are using — method carries marks even when the answer slips.

Easy

EASY · 10 questions · One line from the table. 1 mark each.

1 Differentiate $y = 5\sin x$.

2 Differentiate $y = \cos 2x$.

3 Differentiate $y = e^{5x}$.

4 Differentiate $y = \ln 3x$.

5 Differentiate $y = \tan 4x$.

6 Differentiate $y = x + \sin x$.

7 Differentiate $y = 3e^x - 2 \ln x$.

8 Differentiate $y = \sin(x + 2)$.

9 Differentiate $y = e^{-x}$.

10 Differentiate $y = 4\cos x + \tan x$.

Medium

MEDIUM 10 questions

A chain, a product or a quotient. 2 marks each.

1 Differentiate $y = \cos^3 x$.

2 Differentiate $y = x^2 e^x$.

3 Differentiate $y = \ln(\sin x)$.

4 Differentiate $y = e^{x^2}$.

5 Differentiate $y = \sqrt{\sin x}$.

6 Differentiate $y = \sin x \cdot \cos x$ and simplify.

7 Differentiate $y = \dfrac{e^x}{x}$.

8 Differentiate $y = x \ln x - x$.

9 Differentiate $y = \tan^2 x$.

10 Differentiate $y = (\ln x)^3$.

Hard

HARD 10 questions

Several steps, or an idea. 3 marks each.

1 $y = e^x \cos x$. Find $\frac{d^2y}{dx^2}$.

2 Differentiate $y = \ln \frac{1+x}{1-x}$ and simplify.

3 Differentiate $y = x^{\ln x}$ for $x > 0$.

4 Find the tangent to $y = e^x$ that passes through the origin.

5 Find and classify the stationary point of $y = xe^{-x}$.

6 Show that $y = e^{2x}\sin 3x$ satisfies $y'' - 4y' + 13y = 0$.

7 Differentiate $y = \ln(x + \sqrt{x^2 + 1})$ and simplify.

8 Find the least value of $y = e^x + e^{-x}$.

9 Differentiate $y = \dfrac{1 + \tan x}{1 - \tan x}$ and simplify.

10 Prove that $(\sec x)' = \sec x \cdot \tan x$, starting from $\sec x = \dfrac{1}{\cos x}$.

Answers

Topic 1 · Differentiating sin, cos, tan, e^x and ln x

Easy · easy

1 $5\cos x$

A constant multiple passes straight through the derivative: $5 \cdot \cos x = 5\cos x$.

2 $-2\sin 2x$

Chain rule. The outer derivative is $-\sin 2x$ and the inner derivative is 2, giving $-2\sin 2x$.

3 $5e^{5x}$

The exponential reproduces itself, and the chain rule brings down the inner derivative 5.

4 $\dfrac{1}{x}$

Either use $\dfrac{f'}{f} = \dfrac{3}{3x} = \dfrac{1}{x}$, or note $\ln 3x = \ln 3 + \ln x$, whose derivative is $\dfrac{1}{x}$ since $\ln 3$ is a constant. The 3 disappears either way.

5 $4\sec^2 4x$

The outer derivative is $\sec^2 4x$ and the inner is 4.

6 $1 + \cos x$

Term by term: $(x)' = 1$ and $(\sin x)' = \cos x$.

7 $3e^x - \dfrac{2}{x}$

Term by term, with the constants carried along: $3e^x$ and $-2 \cdot \dfrac{1}{x}$.

8 $\cos(x + 2)$

The inner derivative is 1, so the chain rule contributes nothing beyond the outer derivative.

9 $-e^{-x}$

The inner derivative is -1, so the sign flips. This is the one exponential that decays.

10 $-4\sin x + \sec^2 x$

Term by term. The minus sign belongs to the cosine, not to the 4.

Medium · medium

1 $-3\cos^2 x \cdot \sin x$

The outer function is the cube, giving $3\cos^2 x$; the inner is $\cos x$, whose derivative is $-\sin x$. Multiplying gives $-3\cos^2 x \sin x$.

2 $xe^x(x + 2)$

Product rule: $2x \cdot e^x + x^2 \cdot e^x = e^x(2x + x^2)$, which factorises as $xe^x(x + 2)$.

3 $\cot x$

Use $\dfrac{f'}{f}$ with $f = \sin x$: the derivative is $\dfrac{\cos x}{\sin x} = \cot x$.

4 $2xe^{x^2}$

Chain rule with inner function x^2, whose derivative is $2x$. Note this is not $(e^x)^2$, which would differentiate to $2e^{2x}$.

5 $\dfrac{\cos x}{2\sqrt{\sin x}}$

The outer root gives $\dfrac{1}{2\sqrt{\sin x}}$ and the inner derivative is $\cos x$.

6 $\cos 2x$

Product rule: $\cos x \cdot \cos x + \sin x \cdot (-\sin x) = \cos^2 x - \sin^2 x$, which is $\cos 2x$. Alternatively write $y = \dfrac{\sin 2x}{2}$ first, whose derivative is $\cos 2x$ at once.

7 $\dfrac{e^x(x-1)}{x^2}$

Quotient rule: $\dfrac{e^x \cdot x - e^x \cdot 1}{x^2} = \dfrac{e^x(x-1)}{x^2}$. It is zero at $x = 1$, the minimum of the curve for $x > 0$.

8 $\ln x$

The product gives $\ln x + x \cdot \dfrac{1}{x} = \ln x + 1$, and the $-x$ contributes -1. The two cancel, leaving $\ln x$ — which is why this expression is the antiderivative of $\ln x$.

9 $2 \tan x \cdot \sec^2 x$

The outer square gives $2\tan x$ and the inner derivative is $\sec^2 x$.

10 $\dfrac{3(\ln x)^2}{x}$

The outer cube gives $3(\ln x)^2$ and the inner derivative is $\dfrac{1}{x}$. Keep it apart from $\ln(x^3)$, which is $3\ln x$ and differentiates to $\dfrac{3}{x}$.

Hard · hard

1 $-2e^x \sin x$

First, $y' = e^x \cos x - e^x \sin x = e^x(\cos x - \sin x)$. Differentiating again by the product rule: $e^x(\cos x - \sin x) + e^x(-\sin x - \cos x) = e^x(-2\sin x) = -2e^x \sin x$. The cosine terms cancel exactly.

2 $\dfrac{2}{1-x^2}$

Split the logarithm first: $y = \ln(1+x) - \ln(1-x)$. Then $y' = \dfrac{1}{1+x} + \dfrac{1}{1-x}$, since the second term contributes $-\dfrac{-1}{1-x}$. Over a common denominator this is $\dfrac{(1-x)+(1+x)}{(1+x)(1-x)} = \dfrac{2}{1-x^2}$.

3 $\dfrac{2\ln x}{x} \cdot x^{\ln x}$

Both the base and the index vary, so take logarithms: $\ln y = (\ln x)^2$. Differentiating, $\dfrac{y'}{y} = 2\ln x \cdot \dfrac{1}{x}$, so $y' = x^{\ln x} \cdot \dfrac{2\ln x}{x}$. It is zero at $x = 1$, where the function has its minimum value 1.

4 $y = ex$, touching at $(1, e)$

The tangent at $x = a$ is $y - e^a = e^a(x - a)$, that is $y = e^a x + e^a(1 - a)$. For it to pass through $(0, 0)$ we need $e^a(1 - a) = 0$, and since $e^a \neq 0$ this gives $a = 1$. The tangent is $y = ex$ and it touches at $(1, e)$. Exactly one such tangent exists.

5 Maximum at $x = 1$, of value $\frac{1}{e}$

$y' = e^{-x} - xe^{-x} = e^{-x}(1 - x)$, zero at $x = 1$ since e^{-x} never vanishes. Then $y'' = e^{-x}(x - 2)$, which is $-\frac{1}{e} < 0$ at $x = 1$, so it is a maximum, of value $\frac{1}{e} \approx 0.368$.

6 Verified

$y' = e^{2x}(2\sin 3x + 3\cos 3x)$. Differentiating again, $y'' = 2e^{2x}(2\sin 3x + 3\cos 3x) + e^{2x}(6\cos 3x - 9\sin 3x) = e^{2x}(-5\sin 3x + 12\cos 3x)$. Then $y'' - 4y' + 13y = e^{2x}[(-5 - 8 + 13)\sin 3x + (12 - 12)\cos 3x] = 0$. Both brackets vanish, as required.

7 $\frac{1}{\sqrt{x^2 + 1}}$

Use $\frac{f'}{f}$. The inner derivative is $1 + \frac{x}{\sqrt{x^2 + 1}} = \frac{\sqrt{x^2 + 1} + x}{\sqrt{x^2 + 1}}$. Dividing by $f = x + \sqrt{x^2 + 1}$, that whole bracket cancels and leaves $\frac{1}{\sqrt{x^2 + 1}}$. This function is $\operatorname{arsinh} x$.

8 2, at $x = 0$

$y' = e^x - e^{-x}$, which is zero when $e^x = e^{-x}$, that is $x = 0$. Then $y'' = e^x + e^{-x} = 2 > 0$, a minimum, of value $1 + 1 = 2$. The function is $2\cosh x$, and the result restates that $a + \frac{1}{a} \geq 2$ for positive a.

9 $\frac{2\sec^2 x}{(1 - \tan x)^2}$

Quotient rule: $\frac{\sec^2 x(1 - \tan x) - (1 + \tan x)(-\sec^2 x)}{(1 - \tan x)^2}$. The numerator is $\sec^2 x[(1 - \tan x) + (1 + \tan x)] = 2\sec^2 x$. Since $y = \tan(\frac{\pi}{4} + x)$ by the addition formula, the answer also equals $\sec^2(\frac{\pi}{4} + x)$.

10 $\sec x \cdot \tan x$

By the reciprocal rule, $\frac{d}{dx}\frac{1}{\cos x} = -\frac{(\cos x)'}{\cos^2 x} = \frac{\sin x}{\cos^2 x}$. Split that as $\frac{1}{\cos x} \cdot \frac{\sin x}{\cos x} = \sec x \cdot \tan x$. The two minus signs — one from the rule, one from the derivative of the cosine — cancel, which is why this derivative has no minus sign while $(\operatorname{cosec} x)'$ does.